Charged-Current Weak Interactions

Part (a) — Spin-Summed $|\mathcal{M}|^2$ at High Energy

Equivalent of Griffiths Eq. (9.7) without Mass Terms in the Traces

PHYS 7502 — Particles and Nuclei 2

Problem Statement

(a) Starting with the Feynman Rules for W boson ("Charged Current") interactions (but working at high energy so you can immediately drop *all* mass terms but the W or Z masses) obtain the equivalent of Griffiths Eq. (9.7) without any mass terms inside the traces. **Do NOT go further and evaluate the traces.**

Deliverables

The solution to part (a) will produce, in this order:

1. A clean statement of the Feynman rules used: $W^{\pm}$ vertex factor, $W^{\pm}$ propagator, external fermion factors (Griffiths Eqs. 9.3–9.5).
2. The unsquared invariant amplitude $\mathcal{M}$ for the reference charged-current process, before squaring or spin sums.
3. The pointwise modulus squared $|\mathcal{M}|^2 = \mathcal{M}\mathcal{M}^*$, with explicit Hermitian conjugation of the V−A current.
4. Application of Casimir's trick to convert spinor bilinears into traces *at high energy*, i.e. with the substitution $\sum_s u(p,s)\bar{u}(p,s) \to \not{p}$ (no mc term).
5. The final spin-summed expression, written as a product of two Dirac traces **with no mass terms inside the traces** — the analog of Griffiths Eq. (9.7).

Stopping condition: the result is left as the product of two unevaluated traces.

Background & Setup

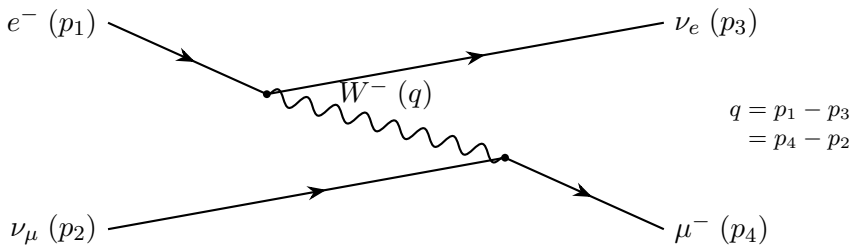
Where this lives. This calculation is the W-boson analogue of the QED calculations in Griffiths Ch. 7 (Casimir's trick, trace theorems). The relevant primary reading is:

- **Griffiths (2nd ed.)**, Ch. 9 *Weak Interactions*, §9.1 *Charged Leptonic Weak Interactions* — Eqs. (9.3)–(9.7), Example 9.1 (Inverse Muon Decay).
- **Thomson**, *Modern Particle Physics*, §11.3–11.5 *Weak charged-current*, §12.2 *Neutrino scattering*.
- **Halzen & Martin**, Ch. 12 *Weak Interactions*, especially §12.1–12.3 for the same trace structures.
- **Mann**, *An Introduction to Particle Physics and the Standard Model*, Ch. 22.

Reference process. We use the same process Griffiths uses to write Eq. (9.7) in Example 9.1 — *inverse muon decay* (often called *neutrino–electron scattering* in Griffiths):

$$\nu_\mu(p_2) + e^-(p_1) \longrightarrow \mu^-(p_4) + \nu_e(p_3),$$

with Griffiths' particle labeling $\{1,2,3,4\} = \{e^-, \nu_\mu, \nu_e, \mu^-\}$. The result will have the same structure for any other tree-level charged-current process; only the labels change.



Concepts & Definitions

Feynman rules for charged-current weak interactions (Griffiths §9.1, Eqs. 9.3–9.5):

- $W^\pm$ **vertex factor** (V−A coupling, Griffiths Eq. (9.5)):

$$-i\,\frac{g_w}{2\sqrt{2}}\,\gamma^\mu\big(1-\gamma^5\big).$$

  g_w is the weak coupling constant; the factor $\frac{1}{2}(1-\gamma^5) \equiv P_L$ is the left-chiral projector (parity is maximally violated because the vertex is V−A, i.e. vector *minus* axial vector).

- $W^\pm$ **propagator** (Griffiths Eq. (9.3)):

$$\frac{-i\big[\,g_{\mu\nu} - q_\mu q_\nu/(M_W c)^2\,\big]}{q^2 - (M_W c)^2}.$$

  At the energies typical of low-energy weak processes ($q^2 \ll (M_W c)^2$), Griffiths Eq. (9.4) reduces this to the simplified form

$$\frac{i\,g_{\mu\nu}}{(M_W c)^2}.$$

- **External fermions:** incoming particle → $u(p,s)$, outgoing particle → $\bar{u}(p,s)$, incoming antiparticle → $\bar{v}(p,s)$, outgoing antiparticle → $v(p,s)$ (Griffiths §7.5).
- **Casimir's trick** (Griffiths Eq. (7.125)): when summing over the spins of an external fermion line, use the completeness relations

$$\sum_s u(p,s)\,\bar{u}(p,s) = \not{p} + mc, \qquad \sum_s v(p,s)\,\bar{v}(p,s) = \not{p} - mc.$$

  The bilinear $\bar{u}\,\Gamma\,u$ becomes a Dirac trace.

Intuition

The W-vertex differs from the QED vertex $-ig_e\gamma^\mu$ in two ways: (i) the coupling strength is $g_w/(2\sqrt{2})$ rather than g_e, and (ii) the Dirac structure is $\gamma^\mu(1-\gamma^5)$ rather than γ^μ. The $(1-\gamma^5)$ projects out only *left-handed* chiral particle states (and right-handed antiparticle states) at every vertex — this is what makes the weak charged current parity-violating. Otherwise, the trace machinery is identical to QED.

Strategy

There are two natural attack routes; both should give the same answer.

Strategy A — Top-down derivation from Feynman rules (recommended):

S1. Write down $i\mathcal{M}$ by reading off the Feynman diagram: two vertices × propagator × four external spinors.

S2. Justify dropping the $q_\mu q_\nu/(M_W c)^2$ piece of the propagator at high energy: when contracted with the external currents, it gives a contribution suppressed by m_f^2/M_W^2 via the Dirac equation, which vanishes when fermion masses are neglected. Use $g_{\mu\nu}/(M_W c)^2$ alone.

S3. Form $|\mathcal{M}|^2 = \mathcal{M}\,\mathcal{M}^*$. The complex conjugate of a current $\bar{u}(b)\gamma^\mu(1-\gamma^5)u(a)$ is $\bar{u}(a)\gamma^\mu(1-\gamma^5)u(b)$ — verify this by computing $\overline{\Gamma^\mu} \equiv \gamma^0\Gamma^{\mu\dagger}\gamma^0$ and using $\{\gamma^0,\gamma^5\}=0$.

S4. Sum over all external spins. Apply Casimir's trick line by line. *Drop every mc that appears* in $\sum_s u\bar{u}$ — this is exactly the high-energy approximation $E \gg m_f$, equivalent to setting $m_e = m_\mu = m_\nu = 0$.

S5. Recognize the resulting closed loops of Dirac matrices as two independent traces — one for each fermion line of the diagram. Stop here.

Strategy B — Drop masses from Griffiths Eq. (9.7) directly:

- Write down Eq. (9.7) verbatim and set $m_e \to 0$, $m_\mu \to 0$ inside the traces. Quick, but skips the conceptual step of justifying *why* the masses can be dropped.

Sanity check (Strategy C): the structure should match the QED $|\mathcal{M}|^2$ for $e^-\mu^- \to e^-\mu^-$ scattering at high energy (Griffiths Eq. 7.126 with the substitution $\gamma^\mu \to \gamma^\mu(1-\gamma^5)$ at every vertex and $1/q^2 \to 1/(M_W c)^2$).

Step-by-Step Derivation (Strategy A)

Step 1 — Write the amplitude from Feynman rules

Reading the diagram from left to right (electron line on top, muon line on bottom):

- Top vertex (electron line, $e^- \to \nu_e + W^-$): $\bar{u}(p_3)\left[-i\dfrac{g_w}{2\sqrt{2}}\gamma^\mu(1-\gamma^5)\right]u(p_1)$.
- W^- propagator carrying q^μ: $\dfrac{-i\left[\,g_{\mu\nu} - q_\mu q_\nu/(M_W c)^2\,\right]}{q^2-(M_W c)^2}$.
- Bottom vertex (muon line, $\nu_\mu + W^- \to \mu^-$): $\bar{u}(p_4)\left[-i\dfrac{g_w}{2\sqrt{2}}\gamma^\nu(1-\gamma^5)\right]u(p_2)$.

Multiplying together gives $i\mathcal{M}$:

$$\begin{aligned} i\mathcal{M} \;=\; &\left[\bar{u}(3)(-i)\tfrac{g_w}{2\sqrt{2}}\gamma^\mu(1-\gamma^5)u(1)\right]\cdot\frac{-i\left[g_{\mu\nu}-q_\mu q_\nu/(M_W c)^2\right]}{q^2-(M_W c)^2} \\ &\times\left[\bar{u}(4)(-i)\tfrac{g_w}{2\sqrt{2}}\gamma^\nu(1-\gamma^5)u(2)\right]. \end{aligned}$$

Step 2 — Drop the $q_\mu q_\nu/(M_W c)^2$ term (tracking every mc)

The W propagator contains two pieces:

$$\frac{-i\left[g_{\mu\nu}-q_\mu q_\nu/(M_W c)^2\right]}{q^2-(M_W c)^2} = \underbrace{\frac{-i\,g_{\mu\nu}}{q^2-(M_W c)^2}}_{\text{keep}} + \underbrace{\frac{+i\,q_\mu q_\nu/(M_W c)^2}{q^2-(M_W c)^2}}_{\text{about to be dropped}}.$$

Sandwiched between the two currents, the longitudinal piece contributes

$$\mathcal{M}_{\text{long}} \;\propto\; \frac{1}{(M_W c)^2}\,\Big[q_\mu\,\bar{u}(3)\gamma^\mu(1-\gamma^5)u(1)\Big]\Big[q_\nu\,\bar{u}(4)\gamma^\nu(1-\gamma^5)u(2)\Big].$$

Each bracket is a number; analyze them independently.

Top current — contract q_μ with the electron line. At the upper vertex, momentum conservation gives $q = p_1 - p_3$, so $\not{q} = \not{p}_1 - \not{p}_3$:

$$q_\mu\,\bar{u}(3)\gamma^\mu(1-\gamma^5)u(1) \;=\; \bar{u}(3)\,\not{q}\,(1-\gamma^5)\,u(1) \;=\; \bar{u}(3)\,(\not{p}_1 - \not{p}_3)\,(1-\gamma^5)\,u(1).$$

Piece A: handling $\not{p}_1$. I want $\not{p}_1 u(1) = m_e c\,u(1)$, but $\not{p}_1$ sits to the *left* of $(1-\gamma^5)$, not adjacent to $u(1)$. Anticommute it through. Using $\{\gamma^\mu, \gamma^5\} = 0$, hence $\not{p}_1\gamma^5 = -\gamma^5\not{p}_1$:

$$\not{p}_1\,(1-\gamma^5) \;=\; \not{p}_1 - \not{p}_1\gamma^5 \;=\; \not{p}_1 + \gamma^5\not{p}_1 \;=\; (1+\gamma^5)\,\not{p}_1.$$

Note the **flip**: $(1-\gamma^5)$ on the right of $\not{p}_1$ becomes $(1+\gamma^5)$ on the left. Now $\not{p}_1$ is adjacent to $u(1)$:

$$\bar{u}(3)\,\not{p}_1(1-\gamma^5)\,u(1) \;=\; \bar{u}(3)\,(1+\gamma^5)\,\not{p}_1\,u(1) \;=\; m_e c\,\bar{u}(3)(1+\gamma^5)u(1).$$

This piece is **proportional to** $m_e c$.

Piece B: handling $\not{p}_3$. Here $\not{p}_3$ is already adjacent to $\bar{u}(3)$, so use the conjugate Dirac equation $\bar{u}(3)\,\not{p}_3 = m_{\nu_e} c\,\bar{u}(3)$:

$$\bar{u}(3)\,\not{p}_3(1-\gamma^5)\,u(1) \;=\; m_{\nu_e} c\,\bar{u}(3)(1-\gamma^5)u(1).$$

This piece is **proportional to** $m_{\nu_e} c$.

Combining the top current:

$$\boxed{q_\mu\,\bar{u}(3)\gamma^\mu(1-\gamma^5)u(1) \;=\; m_e c\,\bar{u}(3)(1+\gamma^5)u(1) \;-\; m_{\nu_e} c\,\bar{u}(3)(1-\gamma^5)u(1)}$$

Bottom current — contract q_ν with the muon line. By identical algebra at the lower vertex, with $q = p_4 - p_2$ giving $\not{q} = \not{p}_4 - \not{p}_2$:

- For $\not{p}_4$ next to $\bar{u}(4)$: directly use $\bar{u}(4)\not{p}_4 = m_\mu c\,\bar{u}(4)$.
- For $\not{p}_2$ next to $u(2)$: anticommute through $(1-\gamma^5)$ as before, then apply $\not{p}_2\,u(2) = m_{\nu_\mu} c\,u(2)$.

Result:

$$\boxed{q_\nu\,\bar{u}(4)\gamma^\nu(1-\gamma^5)u(2) \;=\; m_\mu c\,\bar{u}(4)(1-\gamma^5)u(2) \;-\; m_{\nu_\mu} c\,\bar{u}(4)(1+\gamma^5)u(2)}$$

What just happened to the m's. *Every* surviving term carries a fermion mass — no piece without an mc factor remains. Multiplying the two currents together for $\mathcal{M}_{\text{long}}$:

$$\mathcal{M}_{\text{long}} \;\propto\; \frac{1}{(M_W c)^2}\,(m_e c)(m_\mu c)\,\big[\text{spinor factors}\big] \;\sim\; \frac{m_e\,m_\mu}{M_W^2}\,\big[\text{spinor factors}\big].$$

(With $m_{\nu_e} = m_{\nu_\mu} = 0$, which Griffiths assumes anyway. Including them gives extra cross-terms $\propto m_e m_{\nu_\mu}$ and $\propto m_\mu m_{\nu_e}$, all equally negligible.)

The relative size of the longitudinal piece is therefore

$$\frac{\mathcal{M}_{\text{long}}}{\mathcal{M}_{g_{\mu\nu}}} \sim \frac{m_e\, m_\mu}{M_W^2} \approx \frac{(0.511\,\text{MeV})(105.7\,\text{MeV})}{(80.4\,\text{GeV})^2} \approx 8\times 10^{-9},$$

utterly negligible *even at low energy*, and it vanishes *identically* at high energy where all fermion masses are dropped. That is the rigorous justification for replacing the propagator by

$$\frac{-i\, g_{\mu\nu}}{q^2 - (M_W c)^2}.$$

Physical Insight

One-sentence reason. The $q_\mu q_\nu$ piece encodes the *longitudinal* polarization of the massive W, and longitudinal modes only couple to fermion currents through the fermion mass (this is the Goldstone-equivalence-theorem statement at tree level). Wherever the fermions are light compared to M_W, longitudinal W exchange is suppressed by m_f^2/M_W^2 at the amplitude level, and by m_f^4/M_W^4 in the cross section. So at high energy with $m_f \to 0$, this piece disappears identically.

Step 3 — Take the low-q^2 form (Griffiths Eq. 9.4)

For the inverse-muon-decay reference process, $|q^2| \ll (M_W c)^2$, so the denominator is well approximated by $-(M_W c)^2$, and the propagator becomes

$$\frac{-i\, g_{\mu\nu}}{-(M_W c)^2} = \frac{i\, g_{\mu\nu}}{(M_W c)^2}.$$

(**Aside.** If we are not in the low-q^2 regime, we should keep the full propagator $-i\, g_{\mu\nu}/[q^2 - (M_W c)^2]$ and an overall factor of $1/[q^2 - (M_W c)^2]^2$ replaces $1/(M_W c)^4$ in $|\mathcal{M}|^2$. Either way, no mass terms enter the traces.)

Step 4 — Collect the i's and constants

Three factors of $-i$ from the vertices and propagator combine: $(-i)\cdot(-i)\cdot(-i) = i$. The $g_w/(2\sqrt{2})$ from each vertex gives $g_w^2/8$. So

$$\boxed{\mathcal{M} = \frac{g_w^2}{8\,(M_W c)^2}\left[\bar{u}(3)\gamma^\mu(1-\gamma^5)u(1)\right]\left[\bar{u}(4)\gamma_\mu(1-\gamma^5)u(2)\right]}$$

This is exactly Griffiths Eq. (9.6).

Step 5 — Compute $\mathcal{M}^*$

Each spinor bilinear is a single complex number; its complex conjugate equals its Hermitian conjugate. Using the standard "bar identity" $[\bar{u}(b)\,\Gamma\,u(a)]^* = \bar{u}(a)\,\overline{\Gamma}\,u(b)$ with $\overline{\Gamma} \equiv \gamma^0\Gamma^\dagger\gamma^0$:

For $\Gamma = \gamma^\mu(1-\gamma^5)$:

$$\Gamma^\dagger = (1-\gamma^5)^\dagger(\gamma^\mu)^\dagger = (1-\gamma^5)\,\gamma^0\gamma^\mu\gamma^0,$$
$$\overline{\Gamma} = \gamma^0\,(1-\gamma^5)\,\gamma^0\gamma^\mu\gamma^0\,\gamma^0 = \gamma^0(1-\gamma^5)\gamma^0\,\gamma^\mu.$$

Use $\{\gamma^0,\gamma^5\}=0$, so $\gamma^0\gamma^5\gamma^0=-\gamma^5(\gamma^0)^2=-\gamma^5$, giving $\gamma^0(1-\gamma^5)\gamma^0=1+\gamma^5$. Then $(1+\gamma^5)\gamma^\mu=\gamma^\mu-\gamma^\mu\gamma^5=\gamma^\mu(1-\gamma^5)$ (again using anticommutation of γ^5 with γ^μ). The remarkable conclusion:

$$\overline{\gamma^\mu(1-\gamma^5)}=\gamma^\mu(1-\gamma^5).$$

The V−A current is its own conjugate. Therefore

$$\mathcal{M}^* \;=\; \frac{g_w^2}{8(M_Wc)^2}\big[\bar u(1)\gamma^\nu(1-\gamma^5)u(3)\big]\big[\bar u(2)\gamma_\nu(1-\gamma^5)u(4)\big],$$

where we have renamed the dummy Lorentz index to ν to avoid clash with $\mathcal{M}$.

Warning / Common Pitfall

The dummy Lorentz index in $\mathcal{M}^*$ *must* be different from the index in $\mathcal{M}$. If you sloppily reuse μ, you will accidentally contract indices that should be summed independently and obtain the wrong tensor structure. (This is identical to the QED situation in Griffiths Eq. 7.124.)

Step 6 — Form $|\mathcal{M}|^2$

$$\begin{aligned}|\mathcal{M}|^2 \;=\; \mathcal{M}\,\mathcal{M}^* \;&=\; \left(\frac{g_w^2}{8(M_Wc)^2}\right)^2\\ &\times \underbrace{\big[\bar u(3)\gamma^\mu(1-\gamma^5)u(1)\big]\big[\bar u(1)\gamma^\nu(1-\gamma^5)u(3)\big]}_{\text{electron line}}\\ &\times \underbrace{\big[\bar u(4)\gamma_\mu(1-\gamma^5)u(2)\big]\big[\bar u(2)\gamma_\nu(1-\gamma^5)u(4)\big]}_{\text{muon line}}.\end{aligned}$$

Notice that the spinors group naturally into two independent fermion lines: $\{1,3\}$ on the electron line, $\{2,4\}$ on the muon line. Each line will become a single Dirac trace.

Step 7 — Spin sum, drop mc everywhere

Sum over all four external spins. Use Casimir's trick at high energy:

$$\sum_{\text{spins}} u(p,s)\bar u(p,s) \;=\; \not p + m\,c \;\xrightarrow{E\gg mc^2}\; \not p.$$

This is the moment where every fermion mass disappears *from inside the traces*. Working on the electron line first, write the spinor indices explicitly:

$$\begin{aligned}&\sum_{\text{spins}}\big[\bar u(3)\gamma^\mu(1-\gamma^5)u(1)\big]\big[\bar u(1)\gamma^\nu(1-\gamma^5)u(3)\big]\\ &\quad=\bar u_a(3)[\gamma^\mu(1-\gamma^5)]_{ab}\,u_b(1)\,\bar u_c(1)\,[\gamma^\nu(1-\gamma^5)]_{cd}\,u_d(3)\\ &\quad=[\gamma^\mu(1-\gamma^5)]_{ab}\underbrace{\Big[\sum_{s_1}u_b(1)\bar u_c(1)\Big]}_{(\not p_1)_{bc}}[\gamma^\nu(1-\gamma^5)]_{cd}\underbrace{\Big[\sum_{s_3}u_d(3)\bar u_a(3)\Big]}_{(\not p_3)_{da}}\\ &\quad=\mathrm{Tr}\big[\gamma^\mu(1-\gamma^5)\not p_1\,\gamma^\nu(1-\gamma^5)\not p_3\big].\end{aligned}$$

The closed sum on indices a,b,c,d is a trace because the spinor indices loop back to a at the end. Identical reasoning on the muon line yields

$$\sum_{\text{spins}}\big[\bar u(4)\gamma_\mu(1-\gamma^5)u(2)\big]\big[\bar u(2)\gamma_\nu(1-\gamma^5)u(4)\big] \;=\; \mathrm{Tr}\big[\gamma_\mu(1-\gamma^5)\not p_2\,\gamma_\nu(1-\gamma^5)\not p_4\big].$$

Step 8 — Assemble the result

Key Result

Equivalent of Griffiths Eq. (9.7) with all fermion masses dropped:

$$\sum_{\text{spins}} |\mathcal{M}|^2 = \left(\frac{g_w^2}{8\,(M_W c)^2}\right)^2 \mathrm{Tr}\big[\gamma^\mu(1-\gamma^5)\,\not{p}_1\,\gamma^\nu(1-\gamma^5)\,\not{p}_3\big] \times \mathrm{Tr}\big[\gamma_\mu(1-\gamma^5)\,\not{p}_2\,\gamma_\nu(1-\gamma^5)\,\not{p}_4\big]$$

The two Lorentz indices μ, ν are summed between the two traces. The momenta inside the traces are organized so that particles $\{1, 3\}$ live on the electron line and $\{2, 4\}$ live on the muon line. **No fermion mass appears anywhere inside the traces.** The W-mass M_W survives only as the propagator denominator outside, exactly as the problem requires.
(**Stopping condition.** The traces are left unevaluated, per the problem statement.)

Physical Insight

Why all the mc terms vanish from inside the traces. The Casimir replacement $\sum u\bar{u} = \not{p} + mc$ is the only place a fermion mass enters this calculation. At high energy, $E \gg m_f c^2$, the rest mass is negligible compared to the kinematic momenta, so $\not{p} + mc \to \not{p}$. The W mass is treated differently because M_W enters through the propagator pole, where it sets the inverse range of the weak interaction; you *cannot* drop M_W without changing the physics qualitatively.
Why two traces and not one. Each closed line of fermion arrows in the diagram generates one trace. Inverse muon decay has two distinct fermion lines (electron-line and muon-line), connected only by a W exchange (a boson, not a fermion line); hence the product of two traces. This is structurally identical to QED $e\mu \to e\mu$ scattering, where one also gets one trace per fermion line.
Why $(1-\gamma^5)$ appears twice in each trace, not once. Each end of a fermion line attaches to a vertex, and each vertex carries a factor of $\gamma^\mu(1-\gamma^5)$. So a fermion line that runs through two vertices picks up two V−A factors. The trace theorems of Griffiths §7.7 will eventually combine the two $(1-\gamma^5)$'s using the identity $(1-\gamma^5)\not{p}(1-\gamma^5) = 2\not{p}(1-\gamma^5)$ — but that is the work for parts (b) and beyond, after the trace machinery is applied.

Intuition

Comparison with QED. If you replace each $\gamma^\mu(1-\gamma^5) \to \gamma^\mu$ and $1/(M_W c)^2 \to 1/q^2$, you recover the spin-summed amplitude squared for high-energy $e^-\mu^- \to e^-\mu^-$ scattering (Griffiths Eq. 7.126 with $m_e = m_\mu = 0$). The whole charged-current calculation *is* the QED calculation with the $V - A$ insertion at every vertex and the photon propagator replaced by a contact term. Once you internalize this parallel, every charged-current process becomes structurally familiar.

Warning / Common Pitfall

The 1/2 overall spin-average factor (for the electron initial spin) has *not* been included here — this expression is the un-averaged $\sum_{\text{spins}} |\mathcal{M}|^2$, which is exactly what Griffiths

Eq. (9.7) is. As Griffiths notes (just after Eq. 9.10), the incoming neutrino has only one spin state because the $(1-\gamma^5)$ projector kills the right-handed component; only the electron's two spin states require averaging, giving an overall $\langle|\mathcal{M}|^2\rangle = \frac{1}{2}\sum|\mathcal{M}|^2$. This $1/2$ is applied later, in the cross-section calculation — not at the trace stage.